Section 7.1 Exercises

Directions: For exercises 1 - 11, fully simplify the expression.

1. $\sin x \cos x \sec x$

2. $\sin(-x)\cos(-x)\csc(-x)$

3. $\tan x\sin x + \sec x\cos^2 x$

4. $\csc x + \cos x\cot(-x)$

5. $\frac{\cot t+\tan t}{\sec(-t)}$

6. $3\sin^3 t\csc t + \cos^2 t + 2\cos(-t)\cos t$

7. $-\tan(-x)\cot(-x)$

8. $\frac{-\sin(-x)\cos x \sec x \csc x \tan x}{\cot x}$

9. $\frac{1+\tan^2\theta}{\csc^2\theta} + \sin^2\theta + \frac{1}{\sec^2\theta}$

10. $\left(\frac{\tan x}{\csc^2 x} + \frac{\tan x}{\sec^2 x}\right)\left(\frac{1+\tan x}{1+\cot x}\right) - \frac{1}{\cos^2 x}$

11. $\frac{1-\cos^2 x}{\tan^2 x} + 2\sin^2 x$

Directions: For exercises 12 - 24, simplify the first trigonometric expression by writing the simplified form in terms of the second expression.

12. $\frac{\tan x+\cot x}{\csc x}$; $\cos x$

13. $\frac{\sec x+\csc x}{1+\tan x}$; $\sin x$

14. $\frac{\cos x}{1+\sin x} + \tan x$; $\cos x$

15. $\frac{1}{\sin x\cos x} - \cot x$; $\cot x$

16. $\frac{1}{1-\cos x} - \frac{\cos x}{1+\cos x}$; $\csc x$

17. $(\sec x + \csc x)(\sin x + \cos x) - 2 - \cot x$; $\tan x$

18. $\frac{1}{\csc x-\sin x}$; $\sec x$ and $\tan x$

19. $\frac{1-\sin x}{1+\sin x} - \frac{1+\sin x}{1-\sin x}$; $\sec x$ and $\tan x$

20. $\tan x$; $\sec x$

21. $\sec x$; $\cot x$

22. $\sec x$; $\sin x$

23. $\cot x$; $\sin x$

24. $\cot x$; $\csc x$

Directions: For exercises 25 - 29, prove the identity.

25. $\cos x - \cos^3 x = \cos x \sin^2 x$

26. $\cos x(\tan x - \sec(-x)) = \sin x - 1$

27. $\frac{1+\sin^2 x}{\cos^2 x} = 1 + 2\tan^2 x$

28. $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$

29. $\cos^2 x - \tan^2 x = 2 - \sin^2 x - \sec^2 x$